

Management of diabetic retinopathy: a systematic review.

CONTEXT: Diabetic retinopathy (DR) is the leading cause of blindness in the working-aged population in the United States. There are many new interventions for DR, but evidence to support their use is uncertain. **OBJECTIVE:** To review the best evidence for primary and secondary intervention in the management of DR, including diabetic macular edema. **EVIDENCE ACQUISITION:** Systematic review of all English-language articles, retrieved using a keyword search of MEDLINE (1966 through May 2007), EMBASE, Cochrane Collaboration, the Association for Research in Vision and Ophthalmology database, and the National Institutes of Health Clinical Trials Database, and followed by manual searches of reference lists of selected major review articles. All English-language randomized controlled trials (RCTs) with more than 12 months of follow-up and meta-analyses were included. Delphi consensus criteria were used to identify well-conducted studies. **EVIDENCE SYNTHESIS:** Forty-four studies (including 3 meta-analyses) met the inclusion criteria. Tight glycemic and blood pressure control reduces the incidence and progression of DR. Pan-retinal laser photocoagulation reduces the risk of moderate and severe visual loss by 50% in patients with severe nonproliferative and proliferative retinopathy. Focal laser photocoagulation reduces the risk of moderate visual loss by 50% to 70% in eyes with macular edema. Early vitrectomy improves visual recovery in patients with proliferative retinopathy and severe vitreous hemorrhage. Intravitreal injections of steroids may be considered in eyes with persistent loss of vision when conventional treatment has failed. There is insufficient evidence for the efficacy or safety of lipid-lowering therapy, medical interventions, or antivascular endothelial growth factors on the incidence or progression of DR. **CONCLUSIONS:** Tight glycemic and blood pressure control remains the cornerstone in the primary prevention of DR. Pan-retinal and focal retinal laser photocoagulation reduces the risk of visual loss in patients with severe DR and macular edema, respectively. There is currently insufficient evidence to recommend routine use of other treatments.